UFUG 2106 : Discrete Mathematics

Chapter 08: Properties of Relations

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Readings before lecture: Pages 487-491;495-498; 502;505-508;510-514;517;519

ACK: Part of the PPTs are built from the PPT slides from Instructor's Companion Website of the textbook.

Outline

1. Relations on Sets
2. Reflexivity, Symmetry, and Transitivity
3. Equivalence Relations
4. Modular Arithmetic with Applications to Cryptography
5. Partial Order Relations

Intended Learning Outcomes

By the end of this chapter, students should be able to:

1. **Define and represent** binary and *n*-ary relations using sets, diagrams, and graphs.
2. **Determine** whether a relation is **reflexive**, **symmetric**, or **transitive** using definitions and counterexamples.
3. **Identify and work with equivalence relations**, including finding **equivalence classes** and understanding their connection to partitions.

Intended Learning Outcomes (Continued)

By the end of this chapter, students should be able to:

4. **Use modular arithmetic** to define congruence relations and apply them in simple cryptographic tasks like the **Caesar cipher**.
5. **Recognize and analyze partial orders**, draw **Hasse diagrams**, and perform **topological sorting** on partially ordered sets.

Intended Learning Outcomes (Continued)

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Relations: Definitions Revisited

Definition

Let A and B be sets. A **relation R from A to B** is a subset of $A \times B$. Given an ordered pair (x, y) in $A \times B$, **x is related to y by R**, written $x\,R\,y$, if, and only if, (x, y) is in R. The set A is called the **domain** of R and the set B is called its **co-domain**.

The notation for a relation R may be written symbolically as follows:

$$\boldsymbol{x\,R\,y} \quad \text{means that} \quad \boldsymbol{(x, y) \in R.}$$

The notation $x \not{R}\, y$ means that x is not related to y by R:

$$\boldsymbol{x\,\not{R}\,y} \quad \text{means that} \quad \boldsymbol{(x, y) \notin R.}$$

A more formal way to refer to the kind of relation is to call it a **binary relation** because it is a subset of a Cartesian product of two sets.

Relations: Definitions Revisited

Define a relation L from **R** to **R** as follows: For all real numbers x and y,

$$x \, L \, y \Leftrightarrow x < y.$$

a. Is $57 \, L \, 53$?

b. Is $(-17) \, L \, (-14)$?

c. Is $143 \, L \, 143$?

d. Is $(-35) \, L \, 1$?

e. Draw the graph of L as a subset of the Cartesian plane $\mathbf{R} \times \mathbf{R}$.

The Inverse of a Relation

If R is a relation from A to B, then a relation R^{-1} from B to A can be defined by interchanging the elements of all the ordered pairs of R.

Definition

Let R be a relation from A to B. Define the inverse relation R^{-1} from B to A as follows:

$$R^{-1} = \{(y, x) \in B \times A \mid (x, y) \in R\}.$$

This definition can be written operationally as follows:

For all $x \in A$ and $y \in B$, $\quad (y, x) \in R^{-1} \quad \Leftrightarrow \quad (x, y) \in R.$

The Inverse of a Relation

Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$, and let R be the "divides" relation from A to B: For every ordered pair $(x, y) \in A \times B$,

$$x \, R \, y \quad \Leftrightarrow \quad x \mid y \quad \textit{x divides y.}$$

a. State explicitly which ordered pairs are in R and R^{-1}, and draw arrow diagrams for R and R^{-1}.

b. Describe R^{-1} in words.

Directed Graph of a Relation

Definition

A **relation on a set** A is a relation from A to A.

When a relation R is defined *on* a set A, the arrow diagram of the relation can be modified so that it becomes a **directed graph**.

Instead of representing A as two separate sets of points, represent A only once, and draw an arrow from each point of A to each related point.

Directed Graph of a Relation

As with an ordinary arrow diagram,

> For all points x and y in A,
>
> $$\text{there is an arrow from } x \text{ to } y \quad \Leftrightarrow \quad x\,R\,y \quad \Leftrightarrow \quad (x, y) \in R.$$

Example 8.1.6 – *Directed Graph of a Relation*

Let $A = \{3, 4, 5, 6, 7, 8\}$ and define a relation R on A as follows:

For every $x, y \in A$, $\quad x \, R \, y \quad \Leftrightarrow \quad 2 \mid (x - y).$

Draw the directed graph of R.

N-*ary Relations and Relational Databases*

A special group of relations, called *n*-ary relations, form the mathematical foundation for relational database theory. Just as a binary relation is a subset of a Cartesian product of two sets, an *n*-ary relation is a subset of a **Cartesian product of *n* sets**.

Definition

Given sets $A_1, A_2, \ldots, A_n$, an ***n*-ary relation** R on $A_1 \times A_2 \times \cdots \times A_n$ is a subset of $A_1 \times A_2 \times \cdots \times A_n$. The special cases of 2-ary, 3-ary, and 4-ary relations are called **binary, ternary**, and **quaternary relations**, respectively.

Example 8.1.7 – *A Simple Database*

The following is a radically simplified version of a database that might be used in a hospital. Let A_1 be a set of positive integers, A_2 a set of alphabetic character strings, A_3 a set of numeric character strings, and A_4 a set of alphabetic character strings. Define a quaternary relation R on $A_1 \times A_2 \times A_3 \times A_4$ as follows:

$(a_1, a_2, a_3, a_4) \in R \Leftrightarrow$ a patient with patient ID number a_1, named a_2, was admitted on date a_3, with primary diagnosis a_4.

Example 8.1.7 – *A Simple Database (cont'd)*

At a particular hospital, this relation might contain the following 4-tuples:

(011985, John Schmidt, 020719, asthma)

(574329, Tak Kurosawa, 011419, pneumonia)

(466581, Mary Lazars, 010319, appendicitis)

(008352, Joan Kaplan, 112419, gastritis)

(011985, John Schmidt, 021719, pneumonia)

(244388, Sarah Wu, 010319, broken leg)

(778400, Jamal Baskers, 122719, appendicitis)

Example 8.1.7 – *A Simple Database (cont'd)*

In discussions of relational databases, the *n*-tuples are normally thought of as being written in tables.

Each row of the table corresponds to one *n*-tuple, and the header for each column gives the descriptive attribute for the elements in the column.

Operations within a database allow the data to be manipulated in many different ways.

Example 8.1.7 – *A Simple Database (cont'd)*

For example, in the database language SQL, if the above database is denoted *S*, the result of the query

SELECT Patient_ID#, Name FROM *S* WHERE

Admission_Date = 010319

would be a list of the ID numbers and names of all patients admitted on 01-03-19:

466581	Mary Lazars
244388	Sarah Wu

Example 8.1.7 – *A Simple Database (cont'd)*

This is obtained by taking the intersection of the set $A_1 \times A_2 \times \{010319\} \times A_4$ with the database and then projecting onto the first two coordinates.

Similarly, SELECT can be used to obtain a list of all admission dates of a given patient. For John Schmidt this list is

02–07–19

02–17–19

Example 8.1.7 – *A Simple Database (cont'd)*

Individual entries in a database can be added, deleted, or updated, and most databases can sort data entries in various ways.

In addition, entire databases can be merged, and the entries common to two databases can be moved to a new database.

Example 8.1.7 – *A Simple Database (cont'd)*

Individual entries in a database can be added, deleted, or updated, and most databases can sort data entries in various ways.

In addition, entire databases can be merged, and the entries common to two databases can be moved to a new database.

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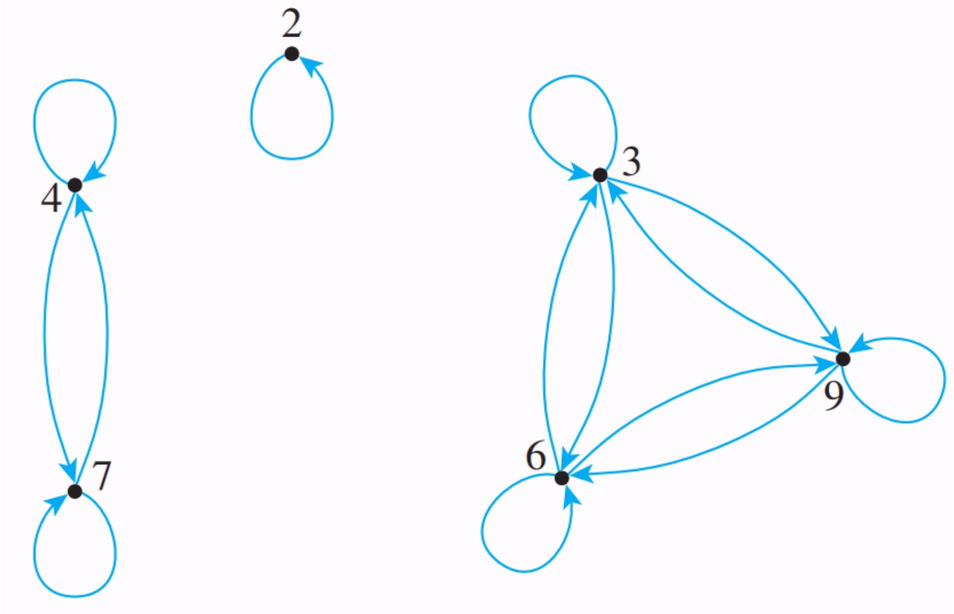
Reflexivity, Symmetry, and Transitivity

Let $A = \{2, 3, 4, 6, 7, 9\}$ and define a relation R on A as follows:

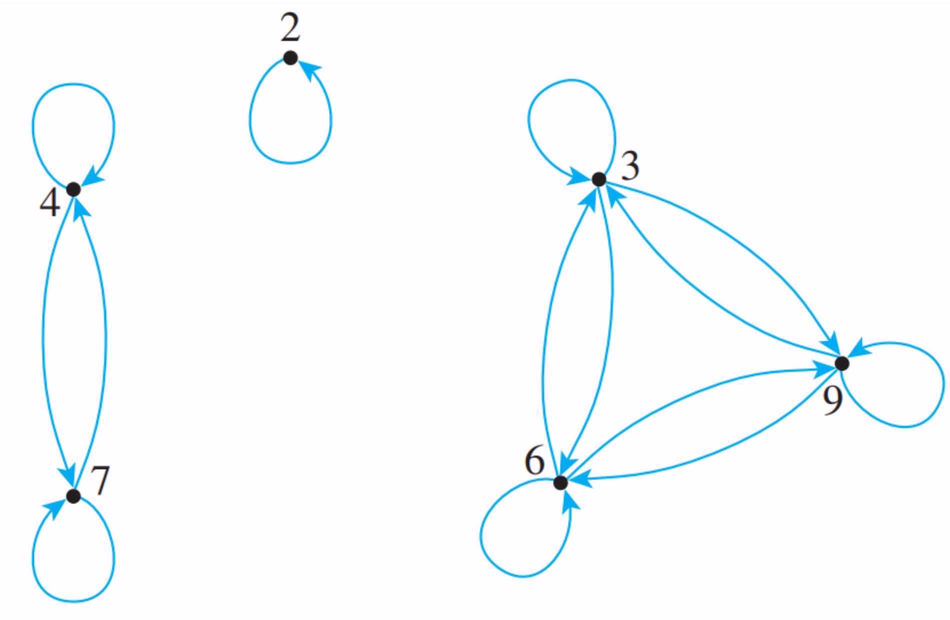
For every $x, y \in A$, $x \; R \; y \Leftrightarrow 3 | (x - y)$



Reflexivity, Symmetry, and Transitivity

This graph has three important properties:

1. Each point of the graph has an arrow looping around from it and going back to it.
2. In each case where there is an arrow going from one point to a second, there is an arrow going from the second point back to the first.
3. In each case where there is an arrow going from one point to a second and from the second point to a third, there is an arrow going from the first point to the third. That is, there are no "incomplete directed triangles" in the graph.



Reflexivity, Symmetry, and Transitivity

Properties (1), (2), and (3) correspond to properties of general relations called *reflexivity*, *symmetry*, and *transitivity*.

Definition

Let R be a relation on a set A.

1. R is **reflexive** if, and only if, for every $x \in A$, $x \, R \, x$.
2. R is **symmetric** if, and only if, for every $x, y \in A$, if $x \, R \, y$ then $y \, R \, x$.
3. R is **transitive** if, and only if, for every $x, y, z \in A$, if $x \, R \, y$ and $y \, R \, z$ then $x \, R \, z$.

Reflexivity, Symmetry, and Transitivity

The equivalence of the expressions $x\ R\ y$ and $(x, y) \in R$ for every x and y in A, the reflexive, symmetric, and transitive properties can also be written as follows:

1. R is reflexive $\Leftrightarrow$ for every x in A, $(x, x) \in R$.
2. R is symmetric $\Leftrightarrow$ for every x and y in A, ***if*** $(x, y) \in R$ then $(y, x) \in R$.
3. R is transitive $\Leftrightarrow$ for every x, y, and z in A, ***if*** $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$.

Reflexivity, Symmetry, and Transitivity

In informal terms, properties (1)–(3) say the following:

Reflexive: Each element is related to itself.

Symmetric: If any one element is related to any other element, then the second element is related to the first.

Transitive: If any one element is related to a second and that second element is related to a third, then the first element is related to the third.

Reflexivity, Symmetry, and Transitivity

Note that the definitions of reflexivity, symmetry, and transitivity are **universal statements**.

This means that to prove a relation has one of the properties, you use either the **method of exhaustion** or the **method of generalizing from the generic particular**.

Reflexivity, Symmetry, and Transitivity

Now consider what it means for a relation **not** to have one of the properties defined previously. Recall that the negation of a universal statement is **existential**. Hence if R is a relation on a set A, then

1. R is **not reflexive** $\Leftrightarrow$ there is an element x in A such that $x \not R x$ *[that is, such that* $(x, x) \notin R$*].*
2. R is **not symmetric** $\Leftrightarrow$ there are elements x and y in A such that $x R y$ but $y \not R x$ *[that is, such that* $(x, y) \in R$ *but* $(y, x) \notin R$*].*
3. R is **not transitive** $\Leftrightarrow$ there are elements x, y, and z in A such that $x R y$ and $y R z$ but $x \not R z$ *[that is, such that* $(x, y) \in R$ *and* $(y, z) \in R$ *but* $(x, z) \notin R$*].*

It follows that you can show that a relation does not have one of the properties by finding a **counterexample**.

Example 8.2.1 – *Properties of Relations on Finite Sets*

Let $A = \{0, 1, 2, 3\}$ and define relations R, S, and T on A as follows:

$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$,

$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\}$,

$T = \{(0, 1), (2, 3)\}$.

a. Is R reflexive? symmetric? transitive?

b. Is S reflexive? symmetric? transitive?

c. Is T reflexive? symmetric? transitive?

Example 8.2.2 – *Properties of Equality*

Define a relation *R* on **R** as follows: For all real numbers *x* and *y*,

$$x \, R \, y \quad \Leftrightarrow \quad x = y.$$

a. Is *R* reflexive? b. Is *R* symmetric? c. Is *R* transitive?

Example 8.2.2 – *Properties of Equality*

Define a relation *R* on **R** as follows: For all real numbers *x* and *y*,

$$\boxed{x \, R \, y \quad \Leftrightarrow \quad x = y.}$$

a. Is *R* reflexive? b. Is *R* symmetric? c. Is *R* transitive?

Example 8.2.3 – *Properties of "Less Than"*

- Define a relation R on $\mathbf{R}$ as follows: For all real numbers x and y,

$$\boxed{x \, R \, y \quad \Leftrightarrow \quad x < y.}$$

a. Is R reflexive? b. Is R symmetric? c. Is R transitive?

The Transitive Closure of a Relation

The relation obtained by adding the least number of ordered pairs to ensure transitivity is called the *transitive closure* of the relation. More precisely, the transitive closure of a relation is the smallest transitive relation that contains the relation.

Definition

Let A be a set and R a relation on A. The **transitive closure** of R is the relation R^t on A that satisfies the following three properties:

1. R^t is transitive.
2. $R \subseteq R^t$.
3. If S is any other transitive relation that contains R, then $R^t \subseteq S$.

Example 8.2.5 – *Transitive Closure of a Relation*

Let $A = \{0, 1, 2, 3\}$ and consider the relation R defined on A as follows:

$$R = \{(0, 1), (1, 2), (2, 3)\}.$$

Find the transitive closure of R.

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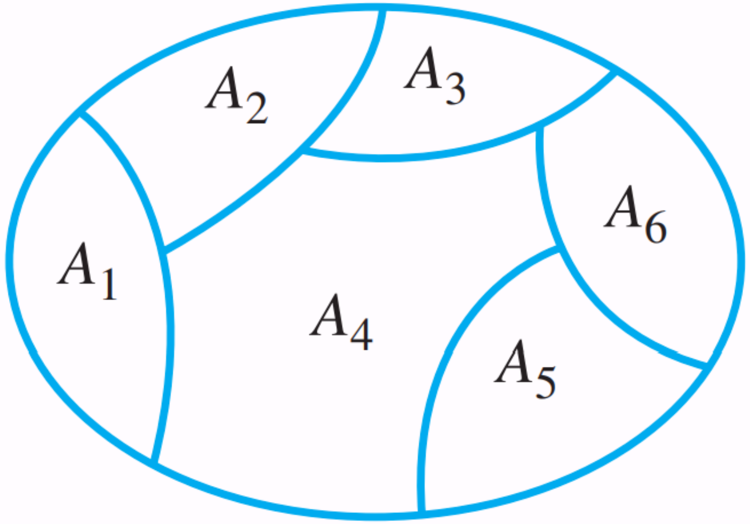
The Relation Induced by a Partition

Definition

Given a partition of a set A, the **relation induced by the partition**, R, is defined on A as follows: For every $x, y \in A$,

$x\,R\,y \quad \Leftrightarrow \quad$ there is a subset A_i of the partition such that both x and y are in A_i.



$$A_i \cap A_j = \emptyset, \text{ whenever } i \neq j$$
$$A_1 \cup A_2 \cup \cdots \cup A_6 = A$$

Example 8.3.1 – *Relation Induced by a Partition*

Let $A = \{0, 1, 2, 3, 4\}$ and consider the following partition of A:

$$\{0, 3, 4\}, \{1\}, \{2\}.$$

Find the relation R induced by this partition.

The Relation Induced by a Partition

Theorem 8.3.1

Let A be a set with a partition and let R be the relation induced by the partition. Then R is reflexive, symmetric, and transitive.

Proof

Definition of an Equivalence Relation

A relation on a set that satisfies the three properties of reflexivity, symmetry, and transitivity is called an *equivalence relation*.

Definition

Let A be a set and R a relation on A. R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

Example 8.3.2 – *An Equivalence Relation on a Set of Subsets*

Let X be the set of all nonempty subsets of {1, 2, 3}. Then

$$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Define a relation **R** on X as follows: For every A and B in X,

A **R** $B \Leftrightarrow$ the least element of A equals the least element of B.

Prove that **R** is an equivalence relation on X.

Equivalence Classes of an Equivalence Relation

Suppose there is an equivalence relation on a certain set. If *a* is any particular element of the set, then one can ask, "What is the subset of all elements that are related to *a*?" This subset is called the *equivalence class* of *a*.

Definition

Suppose A is a set and R is an equivalence relation on A. For each element a in A, the **equivalence class of a**, denoted $[a]$ and called the **class of a** for short, is the set of all elements x in A such that x is related to a by R.

In symbols:

$$[a] = \{x \in A \,|\, x \, R \, a\}$$

Equivalence Classes of an Equivalence Relation

The procedural version of this definition is

$$\boxed{\text{for every } x \in A, \quad x \in [a] \quad \Leftrightarrow \quad x \, R \, a.}$$

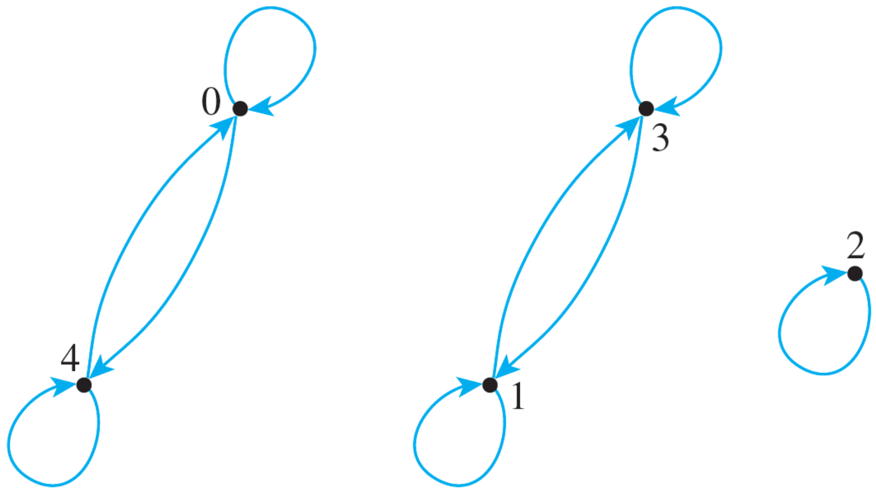
The notation $[a]_R$ may be used to denote the equivalence class of a for the relation R.

Example 8.3.5 – *Equivalence Classes of a Relation Given as a Set of Ordered Pairs*

Let $A = \{0, 1, 2, 3, 4\}$ and define a relation R on A as follows:

$$R = \{(0, 0), (0, 4), (1, 1), (1, 3), (2, 2), (3, 1), (3, 3), (4, 0), (4, 4)\}.$$

R is an equivalence relation on A. Find the distinct equivalence classes of R.



Example 8.3.8 – *Equivalence Classes of the Identity Relation*

Let A be any set and define a relation R on A as follows: For every x and y in A,

$$\boxed{x \, R \, y \quad \Leftrightarrow \quad x = y.}$$

Then R is an equivalence relation. Describe the distinct equivalence classes of R.

Equivalence Classes of an Equivalence Relation

The first lemma says that if two elements of *A* are related by an equivalence relation *R*, then their equivalence classes are the same.

Lemma 8.3.2

Suppose A is a set, R is an equivalence relation on A, and a and b are elements of A. If $a\ R\ b$, then $[a] = [b]$.

Equivalence Classes of an Equivalence Relation

The second lemma says that any two equivalence classes of an equivalence relation are either mutually disjoint or identical.

Lemma 8.3.3

If A is a set, R is an equivalence relation on A, and a and b are elements of A, then

$$\text{either} \quad [a] \cap [b] = \varnothing \quad \text{or} \quad [a] = [b].$$

Equivalence Classes of an Equivalence Relation

Theorem 8.3.4 The Partition Induced by an Equivalence Relation

If A is a set and R is an equivalence relation on A, then the distinct equivalence classes of R form a partition of A; that is, the union of the equivalence classes is all of A, and the intersection of any two distinct classes is empty.

Example 8.3.10 – *Equivalence Classes of Congruence Modulo 3*

Let *R* be the relation of congruence modulo 3 on the set **Z** of all integers. That is, for all integers *m* and *n*,

$$\boxed{m \, R \, n \quad \Leftrightarrow \quad 3 \mid (m - n).}$$

Describe the distinct equivalence classes of *R*.

Congruence Modulo n

Definition

Suppose R is an equivalence relation on a set A and S is an equivalence class of R. A **representative** of the class S is any element a such that $[a] = S$.

Example 8.3.12 – *Rational Numbers Are Really Equivalence Classes*

Let A be the set of all ordered pairs of integers for which the second element of the pair is nonzero. Symbolically: $A = \mathbf{Z} \times (\mathbf{Z} - \{0\})$.

Define a relation R on A as follows: For all pairs (a, b) and (c, d) in A,

$$\boxed{(a, b)\, R\, (c, d) \quad \Leftrightarrow \quad ad = bc.}$$

The fact is that R is an equivalence relation.

1. Prove that R is transitive.
2. Describe the distinct equivalence classes of R.

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Modular Arithmetic with Applications to Cryptography

Cryptography is the study of methods for sending secret messages. It involves **encryption**, in which a message, called **plaintext**, is converted into a form, called **ciphertext**, that is sent over a channel possibly open to view by outside parties.

The receiver of the ciphertext uses **decryption** to convert the ciphertext back into plaintext.

Modular Arithmetic with Applications to Cryptography

Many systems for sending secret messages require both the sender and the receiver to know both the encryption and the decryption procedures.

For instance, an encryption system once used by Julius Caesar, and now called the **Caesar cipher**, encrypts messages by changing each letter of the alphabet to the one three places farther along, with X wrapping around to A, Y to B, and Z to C.

Modular Arithmetic with Applications to Cryptography

In other words, say each letter of the alphabet is coded by its position relative to the others—so that

A = 01, B = 02, ... , Z = 26.

If the numerical version of the plaintext for a letter is denoted *M* and the numeric version of the ciphertext is denoted *C*, then

$$\boxed{C = (M + 3) \; mod \; 26.}$$

Modular Arithmetic with Applications to Cryptography

The receiver of such a message can easily decrypt it by using the formula

$$\boxed{M = (C - 3)\ mod\ 26.}$$

For reference, here are the letters of the alphabet, together with their numeric equivalents:

A 01	B 02	C 03	D 04	E 05	F 06	G 07	H 08	I 09	J 10	K 11	L 12	M 13
N 14	O 15	P 16	Q 17	R 18	S 19	T 20	U 21	V 22	W 23	X 24	Y 25	Z 26

Example 8.4.1 – *Encrypting and Decrypting with the Caesar Cipher*

1. Use the Caesar cipher to encrypt the message HOW ARE YOU.
2. Use the Caesar cipher to decrypt the message L DP ILQH.

A 01	B 02	C 03	D 04	E 05	F 06	G 07	H 08	I 09	J 10	K 11	L 12	M 13
N 14	O 15	P 16	Q 17	R 18	S 19	T 20	U 21	V 22	W 23	X 24	Y 25	Z 26

Properties of Congruence Modulo *n*

Theorem 8.4.1 Modular Equivalences

Let a, b, and n be any integers and suppose $n > 1$. The following statements are all equivalent:

1. $n \mid (a - b)$
2. $a \equiv b \pmod{n}$
3. $a = b + kn$ for some integer k
4. a and b have the same (nonnegative) remainder when divided by n
5. $a \bmod n = b \bmod n$

Properties of Congruence Modulo *n*

Another consequence of the quotient-remainder theorem is this:

When an integer a is divided by a positive integer n, a unique quotient q and remainder r are obtained with the property that $a = nq + r$ and $0 \leq r < n$.

Because there are exactly n integers that satisfy the inequality $0 \leq r < n$ (the numbers from 0 through $n - 1$), there are exactly n possible remainders that can occur.

These are called the least nonnegative residues modulo n or simply the residues modulo n

Properties of Congruence Modulo n

Definition

Given integers a and n with $n > 1$, **the residue of *a* modulo *n*** is $a \bmod n$, the nonnegative remainder obtained when a is divided by n. The numbers $0, 1, 2, \ldots, n-1$ are called a **complete set of residues modulo *n***. To **reduce a number modulo *n*** means to set it equal to its residue modulo n. If a modulus $n > 1$ is fixed throughout a discussion and an integer a is given, the words "modulo n" are often dropped and we simply speak of **the residue of *a***.

Properties of Congruence Modulo *n*

Theorem 8.4.2 Congruence Modulo *n* Is an Equivalence Relation

If n is any integer with $n > 1$, congruence modulo n is an equivalence relation on the set of all integers. The distinct equivalence classes of the relation are the sets [0], [1], [2], . . . , $[n-1]$, where for each $a = 0, 1, 2, \ldots, n-1$,

$$[a] = \{m \in Z \mid m \equiv a \pmod{n}\},$$

or, equivalently,

$$[a] = \{m \in Z \mid m = a + kn \text{ for some integer } k\}.$$

Properties of Congruence Modulo n

Theorem 8.4.2 Congruence Modulo n Is an Equivalence Relation

If n is any integer with $n > 1$, congruence modulo n is an equivalence relation on the set of all integers. The distinct equivalence classes of the relation are the sets $[0]$, $[1]$, $[2]$, . . . , $[n-1]$, where for each $a = 0, 1, 2, \ldots, n-1$,

$$[a] = \{m \in Z \mid m \equiv a \pmod{n}\},$$

or, equivalently,

$$[a] = \{m \in Z \mid m = a + kn \text{ for some integer } k\}.$$

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Antisymmetry

In earlier section, we have defined three properties of relations: **reflexivity**, **symmetry**, and **transitivity**. A fourth property of relations is called ***antisymmetry***.

In terms of the arrow diagram of a relation, saying that a relation is antisymmetric is the same as saying that whenever there is an arrow going from one element to another *distinct* element, there is *not* an arrow going back from the second to the first.

Antisymmetry

Definition

Let R be a relation on a set A. R is **antisymmetric** if, and only if,

$$\text{for every } a \text{ and } b \text{ in } A, \quad \text{if } a\,R\,b \text{ and } b\,R\,a \text{ then } a = b.$$

By taking the negation of the definition, you can see that a relation R is ***not* antisymmetric** if, and only if,

there are elements a and b in A such that $a\,R\,b$ and $b\,R\,a$ but $a \neq b$.

Example 8.5.1 – *Testing for Antisymmetry of Finite Relations*

Let R_1 and R_2 be the relations on {0, 1, 2} defined as follows: Draw the directed graphs for R_1 and R_2 and indicate which relations are antisymmetric.

a. $R_1 = \{(0, 2), (1, 2), (2, 0)\}$

b. $R_2 = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2)\}$

Example 8.5.2 – *Testing for Antisymmetry of "Divides" Relations*

Let R_1 be the "divides" relation on the set of all positive integers, and let R_2 be the "divides" relation on the set of all integers.

For every $a, b \in Z^+$, $\quad a\,R_1\,b \;\Leftrightarrow\; a \mid b.$

For every $a, b \in Z$, $\quad a\,R_2\,b \;\Leftrightarrow\; a \mid b.$

a. Is R_1 antisymmetric? Prove or give a counterexample.

b. Is R_2 antisymmetric? Prove or give a counterexample.

Partial Order Relations

A relation that is reflexive, antisymmetric, and transitive is called a *partial order*.

Definition

Let R be a relation defined on a set A. R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

Two fundamental partial order relations are the "less than or equal to" relation on a set of real numbers and the "subset" relation on a set of sets. These can be thought of as models, or paradigms, for general partial order relations.

Example 8.5.3 – *The "Subset" Relation*

Let A be any collection of sets and define the "subset" relation, ⊆, on A as follows:

For every $U, V \in A$, $U \subseteq V \Leftrightarrow \text{for each } x, \text{if } x \in U \text{ then } x \in V$.

We know that ⊆ is reflexive and transitive. Finish the proof that ⊆ is a partial order relation by proving that ⊆ is antisymmetric.

Example 8.5.4 – *A "Divides" Relation on a Set of Positive Integers*

Let | be the "divides" relation on a set *A* of positive integers. That is, for all *a* and *b* in *A*,

$$a \mid b \quad \Leftrightarrow \quad b = ka \text{ for some integer } k.$$

Prove that | is a partial order relation on *A*.

Partial Order Relations

Notation

Because of the special paradigmatic role played by the $\leq$ relation in the study of partial order relations, the symbol $\preceq$ is often used to refer to a general partial order relation, and the notation $x \preceq y$ is read "x is less than or equal to y" or "y is greater than or equal to x."

Lexicographic Order

Theorem 8.5.1 Lexicographic Order

Let A be a set with a partial order relation R, and let S be a set of strings over A. Define a relation $\preceq$ on S as follows:

Let s and t be any strings in S of lengths m and n, respectively, where m and n are positive integers, and let s_m and t_m be the characters in the mth position for s and t, respectively.

1. If $m \leq n$ and the first m characters of s and t are the same, then $s \preceq t$.
2. If the first $m - 1$ characters in s and t are the same, $s_m \, R \, t_m$, and $s_m \neq t_m$, then $s \preceq t$.
3. If λ is the null string then $\lambda \preceq s$.

If no strings are related by $\preceq$ other than by these three conditions, then $\preceq$ is a partial order relation on S.

Definition

The partial order relation of Theorem 8.5.1 is called the **lexicographic order for S** that corresponds to the partial order R on A.

Example 8.5.6 – *Testing Strings for Lexicographic Order*

Let $A = \{x, y\}$ and let R be the following partial order relation on A: $R = \{(x, x), (x, y), (y, y)\}$.

Let S be the set of all strings over A, and denote by the lexicographic order for S that corresponds to R.

a. Is $x \preceq x$? Is $x \preceq xx$? Is $yx \preceq yxy$?

b. Is $xxxyyy \preceq xy$?

c. Is $x \preceq y$?

d. Is $\lambda \preceq xy$?

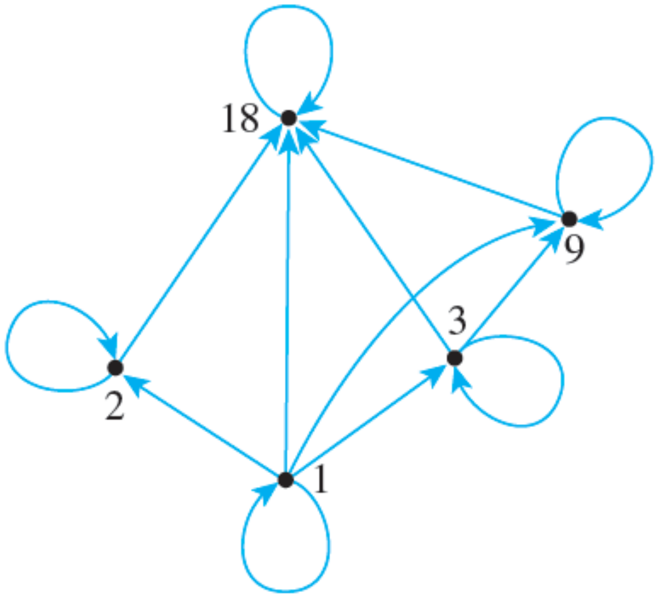
e. Is $xyy \preceq xyx$?

Hasse Diagrams

Let A = {1, 2, 3, 9, 18} and consider the "divides" relation on A: For every a, $b \in A$,

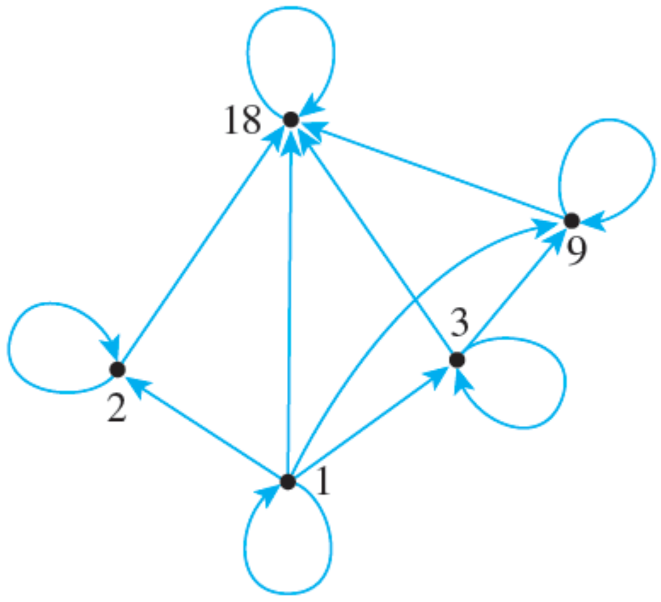
$$\boxed{a \mid b \quad \Leftrightarrow \quad b = ka \text{ for some integer } k.}$$

The directed graph of this relation has the following appearance:



Hasse Diagrams

Note that there is a loop at every vertex, all other arrows point in the same direction (upward), and any time there is an arrow from one point to a second and from the second point to a third, there is an arrow from the first point to the third.



Given any partial order relation defined on a finite set, it is possible to draw the directed graph in such a way that all of these properties are satisfied.

Hasse Diagrams

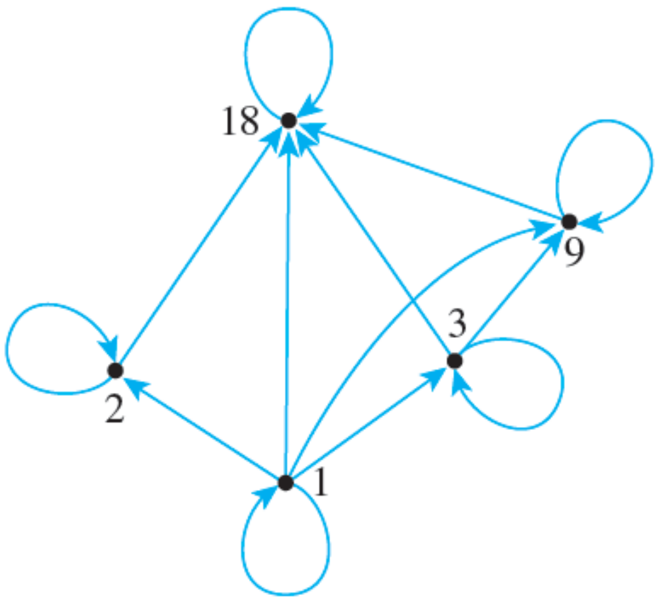
This makes it possible to associate a somewhat simpler graph, called a **Hasse diagram**, with a partial order relation defined on a finite set.

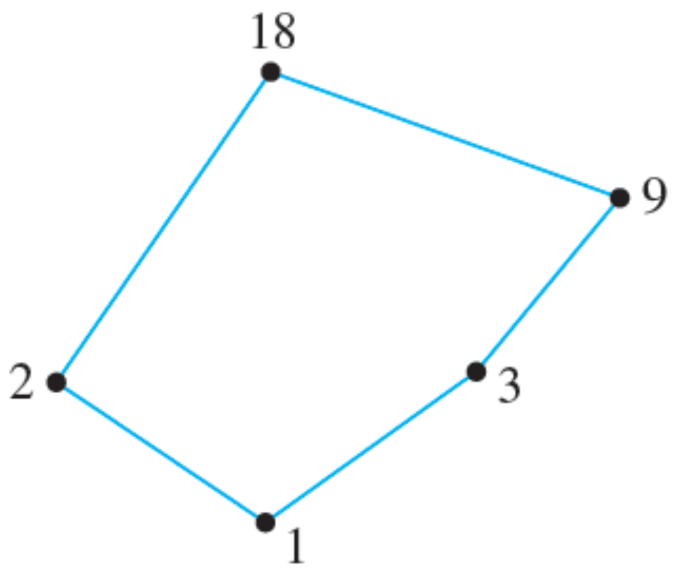
To obtain a Hasse diagram, proceed as follows: Start with a directed graph of the relation, placing vertices on the page so that all arrows point upward.

Then eliminate

1. the loops at all the vertices,
2. all arrows whose existence is implied by the transitive property, and
3. the direction indicators on the arrows.

Hasse Diagrams





Example 8.5.7 – *Constructing a Hasse Diagram*

Consider the "subset" relation, ⊆, on the set $P(\{a, b, c\})$.

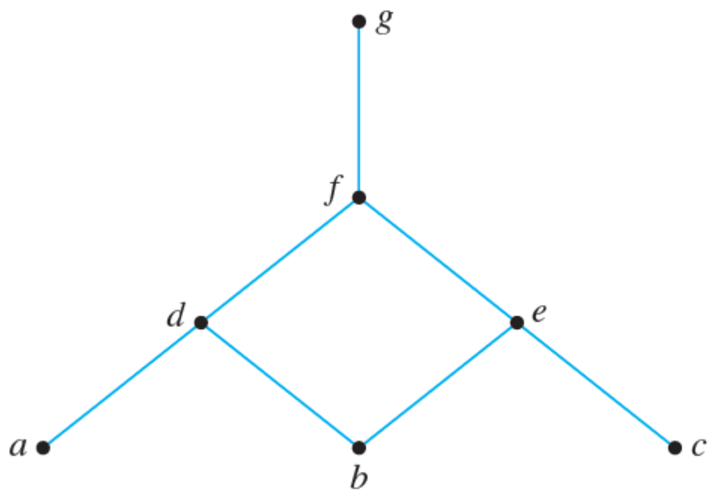
That is, for all sets *U* and *V* in $P(\{a,b,c\})$,

$$\boxed{U \subseteq V \quad \Leftrightarrow \quad \forall x, \text{if } x \in U \text{ then } x \in V.}$$

Construct the Hasse diagram for this relation.

Example 8.5.8 – *Obtaining the Directed Graph of a Partial Order Relation from the Hasse Diagram of the Relation*

A partial order relation *R* has the following Hasse diagram. Find the directed graph of *R*.



Partially and Totally Ordered Sets

Given any two real numbers x and y, either $x \leq y$ or $y \leq x$. In a situation like this, the elements x and y are said to be ***comparable***. On the other hand, given two subsets A and B of $\{a, b, c\}$, it may be the case that neither $A \subseteq B$ nor $B \subseteq A$. For instance, let $A = \{a, b\}$ and $B = \{b, c\}$. Then $A \nsubseteq B$ and $B \nsubseteq A$. In such a case, A and B are said to be ***noncomparable***.

Definition

Suppose $\preceq$ is a partial order relation on a set A. Elements a and b of A are said to be **comparable** if, and only if, either $a \preceq b$ or $b \preceq a$. Otherwise, a and b are called **noncomparable**.

Partially and Totally Ordered Sets

When all the elements of a partial order relation are comparable, the relation is called a ***total order***.

Definition

If R is a partial order relation on a set A, and for any two elements a and b in A either $a \; R \; b$ or $b \; R \; a$, then R is a **total order relation** on A.

Both the "less than or equal to" relation on sets of real numbers and the lexicographic order of the set of words in a dictionary are total order relations. Note that the Hasse diagram for a total order relation can be drawn as a single vertical "chain."

Partially and Totally Ordered Sets

A set A is called a **partially ordered set** (or **poset**) with respect to a relation $\preccurlyeq$, if and only if, $\preccurlyeq$ is a partial order relation on A.

For instance, the set of real numbers is a partially ordered set with respect to the "less than or equal to" relation $\leq$, and a set of sets is partially ordered with respect to the "subset" relation $\subseteq$.

Partially and Totally Ordered Sets

It is entirely straightforward to show that any subset of a partially ordered set is partially ordered (left as exercise).

This, of course, assumes the “same definition” for the relation on the subset as for the set as a whole. A set A is called a totally ordered set with respect to a relation $\preccurlyeq$ if, and only if, A is partially ordered with respect to $\preccurlyeq$ and $\preccurlyeq$ is a total order.

Chain

Definition

Let A be a set that is partially ordered with respect to a relation $\preceq$. A subset B of A is called a **chain** if, and only if, the elements in each pair of elements in B are comparable. In other words, $a \preceq b$ or $b \preceq a$ for every a and b in B. The **length of a chain** is one less than the number of elements in the chain.

Example 8.5.9 – *A Chain of Subsets*

The set $P(\{a, b, c\})$ is partially ordered with respect to the subset relation. Find a chain of length 3 in $P(\{a, b, c\})$.

Partially and Totally Ordered Sets

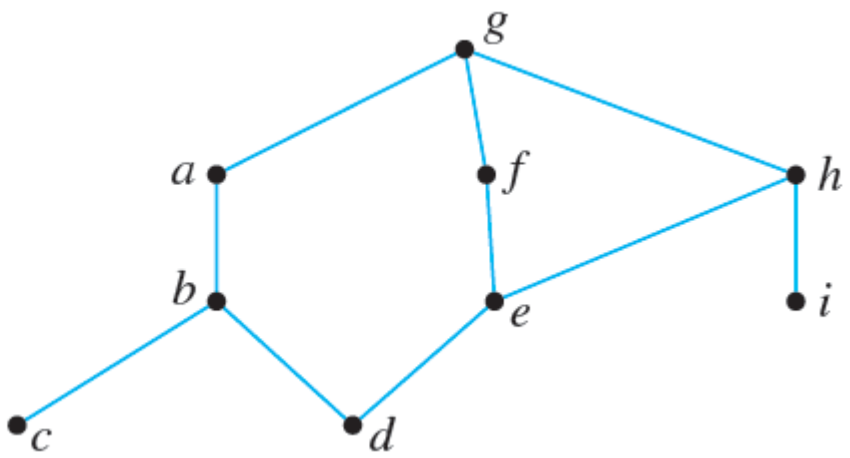
Definition

Let a set A be partially ordered with respect to a relation $\preceq$.

1. An element a in A is called a **maximal element of *A*** if, and only if, for each b in A, either $b \preceq a$ or b and a are not comparable.
2. An element a in A is called a **greatest element of *A*** if, and only if, for each b in A, $b \preceq a$.
3. An element a in A is called a **minimal element of *A*** if, and only if, for each b in A, either $a \preceq b$ or b and a are not comparable.
4. An element a in A is called a **least element of *A*** if, and only if, for each b in A, $a \preceq b$.

Example 8.5.10 – *Maximal, Minimal, Greatest, and Least Elements*

Let $A = \{a, b, c, d, e, f, g, h, i\}$ have the partial ordering $\preccurlyeq$ defined by the following Hasse diagram.



Find all maximal, minimal, greatest, and least elements of A.

Topological Sorting

Definition

Given partial order relations $\preceq$ and $\preceq'$ on a set A, $\preceq'$ is **compatible** with $\preceq$ if, and only if, for every a and b in A, if $a \preceq b$ then $a \preceq' b$.

A total order that is compatible with a given order is called a *topological sorting*.

Definition

Given partial order relations $\preceq$ and $\preceq'$ on a set A, $\preceq'$ is a **topological sorting** for $\preceq$ if, and only if, $\preceq'$ is a total order that is compatible with $\preceq$.

Topological Sorting

Constructing a Topological Sorting

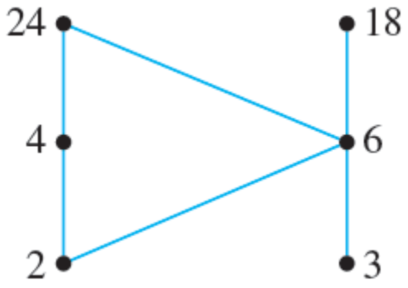
Let $\preceq$ be a partial order relation on a nonempty finite set A. To construct a topological sorting:

1. Pick any minimal element x in A. *[Such an element exists since A is nonempty.]*
2. Set $A' := A - \{x\}$.
3. Repeat steps a–c while $A' \neq \emptyset$.
   a. Pick any minimal element y in A'.
   b. Define $x \preceq' y$.
   c. Set $A' := A' - \{y\}$ and $x := y$.

   [Completion of steps 1–3 of this algorithm gives enough information to construct the Hasse diagram for the total ordering $\preceq$. We have already shown how to use the Hasse diagram to obtain a complete directed graph for a relation.]

Example 8.5.11 – *A Topological Sorting*

Consider the set $A = \{2, 3, 4, 6, 18, 24\}$ ordered by the "divides" relation $|$. The Hasse diagram of this relation is the following:



The ordinary "less than or equal to" relation $\leq$ on this set is a topological sorting for it since for positive integers *a* and *b*, if $a|b$ then $a<b$. Find another topological sorting for this set.